



MATSIM USER MEETING 2026 · PARIS · 28 SEPTEMBER

CitySim • **AI**

One Data Pipeline, Three Cities

*for pedestrian safety, flood resilience and
event traffic management*

Port of Spain · Varanasi · Miami-Dade

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CityData • **AI**

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DYNAMIC DIGITAL TWINS OF CITIES

people counts · density patterns · movement trends



Commercial Districts

for economic development



Mobility & Transportation

for transportation agencies



Parks & Open Spaces

for parks & recreation

Raw Data Sources

The hard part isn't running the simulation. It's the demand.



curates over

10,000+

mobility data sources

*Anonymised movement patterns, not survey responses:
OD flows, route trajectories, trip-level speeds, distances
and travel times.*

35–65%

of the metro population

6 years

of data, every city worldwide

CityFlow AI

CityEconomy AI

CityParks AI

CitySensor AI

MOBILE APPS

Location-enabled apps: games, news, entertainment, productivity

CONNECTED VEHICLES

Telemetry from fleets of connected vehicles

WIFI HOTSPOTS

Open access points for public use

COUNTING BEACONS

Bluetooth beacons in malls and commercial zones

IOT SENSORS

Camera sensors, traffic counters, charging stations

From Data to a Plans File

Fusion, calibration – and where CitySim draws the line

CityFlow•AI

Passive mobility data: OD flows, routes, trip-level travel times

CityEconomy•AI

Visits, spending and visitor demographics: the core patterns of life

CityParks•AI

Visits and movement in parks, open spaces and nature preserves

CitySensor•AI

Field hardware measuring ground truth

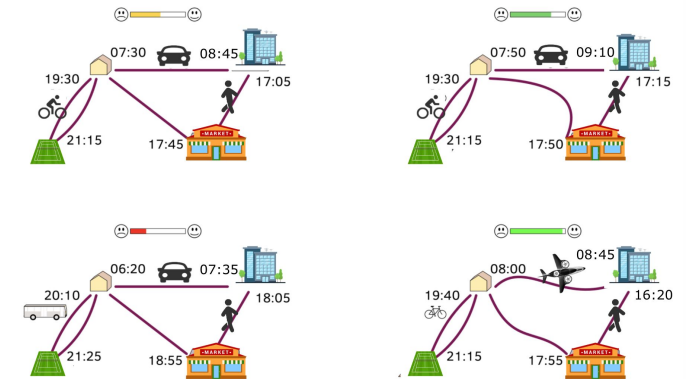


fuses and calibrates the inputs into a realistic population and travel demand

PROPRIETARY ←

→ OPEN & REPRODUCIBLE

plans file



MATSim plans: a full day of activities and trips for every agent

We're deliberately transparent about that line – academically and commercially.



THE HYPOTHESIS

**If we can industrialise the difficult demand-generation step,
then deploying a city simulation stops being a bespoke
research project every time and starts becoming a
repeatable process.**

*We've tested that hypothesis across three cities, with three very different problems.
And in each case, MATSim ends up doing a different job.*

CASE STUDY 1

Port of Spain

MATSim as scenario comparator

Does a flood's disruption stay local – or propagate through the network?

Two otherwise identical scenarios: the normal network, and the same day with the flood-prone roads removed.

1

Hand-drawn KMZ

Not a clean list of road IDs

2

Matched to the network

Those lines onto real road links

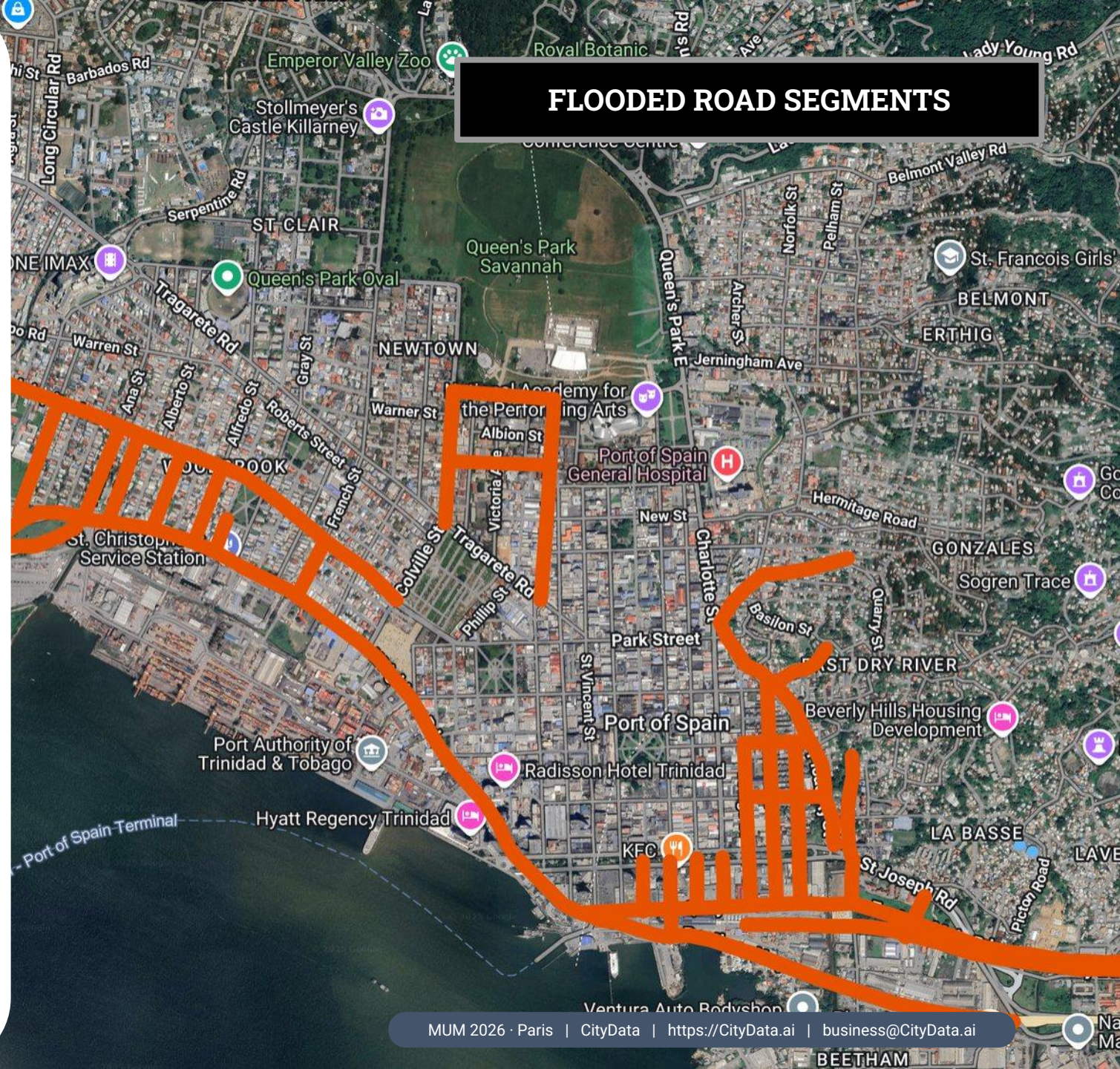
3

Removed & rebuilt

Roughly 1 in 60 links; bus routes remapped

Asked for by the UNDP team in Trinidad and Tobago.

FLOODED ROAD SEGMENTS



PORT OF SPAIN · THE RESULT

+1%

total travel distance

+41%

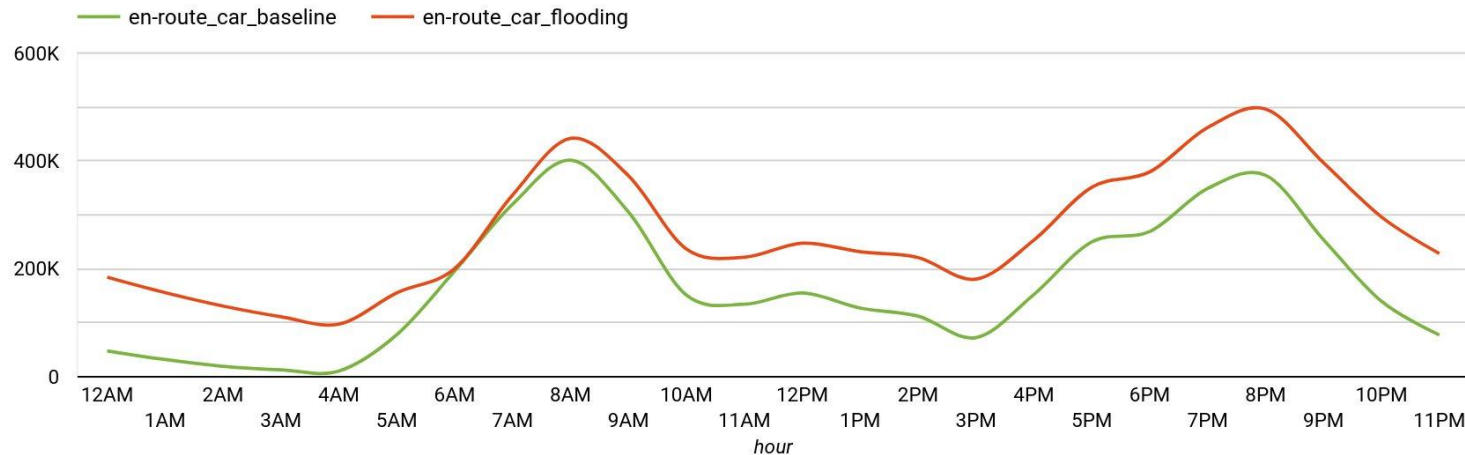
travel time

People weren't travelling dramatically farther.
They were spending much more time stuck in congestion.

The Failure to Recover

PORT OF SPAIN

The result that interested us more than the travel time



Vehicles en route through the day – baseline vs flooding · CitySim dashboard

On a normal day, morning congestion clears by early afternoon. Under the flood scenario, it doesn't.

278,000 flooded

216,000 normal day

vehicles still en route as the evening peak arrives

Which parts of the network make the system vulnerable, and how quickly does it recover when they fail?

A more useful question than "flooding causes delay" – and one planners can act on.

CASE STUDY 2

Varanasi

MATSim as forecasting engine

Can we provide several hours of warning that a dangerous crowd might be developing?

~2 m

lane widths in the historic core

60 M+

pilgrims in peak festive season

8

CitySensors installed in the study area

Crowd crushes are a real safety concern — and the city has very limited visibility.

GODOWLIA CHAURAHA



Forecasting With the Simulation

VARANASI

Not “400 now, so 450 later.”



Simulated pedestrian flows through the historic core

1

Simulate ordinary days

CityFlow data → day plans → pedestrians on the real street network, minute by minute

2

Watch the live day

The available cameras, from midnight onwards

3

Match, then carry forward

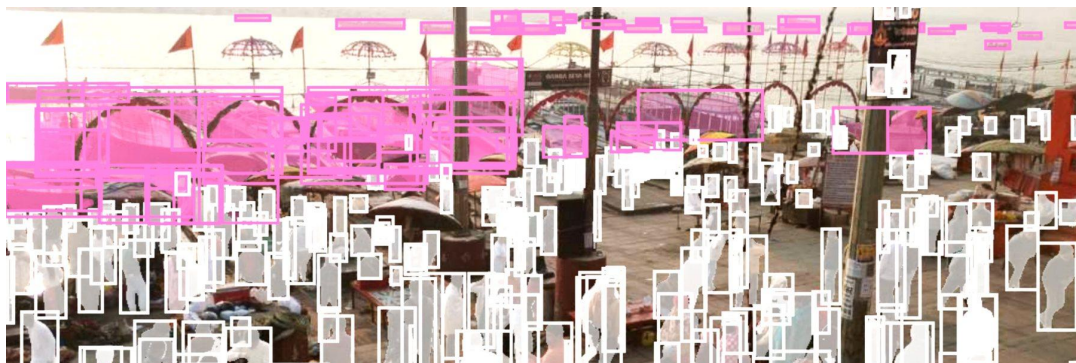
Which of those days looks most like today so far? Fit it, and let the simulation run on

A camera tells you how many people are at one location.

It doesn't tell you where they're going next, which route they'll take, or which narrow lane is about to overload.

Does It Work?

A useful predictive signal – and three things we won't claim



MAE $\approx$ 23 people

on counts from near zero to several hundred

The open research question

Can a physically consistent simulation turn a few sensor measurements into a picture of what's happening across the whole network?

VARANASI

THREE THINGS WE WON'T CLAIM

Only ~1 month of simulated days

Enough to demonstrate the concept – nowhere near enough for festivals, unusual events, or severe weather.

Accuracy drops later in the day

Even with more observations available. We don't yet know why.

The simpler method still wins

Our camera-count approach currently beats the simulation on raw forecast error. Not a solved problem.

Toyota Mobility Foundation award

Continuing the Varanasi research, and expanding to 50 CitySensor sites over the next 12 months.

CASE STUDY 3



Miami-Dade

MATSim as intervention testbed

Can we test traffic-management and egress plans before the event, not during it?

TWO CHALLENGES

1 Traffic signals

NETWORK

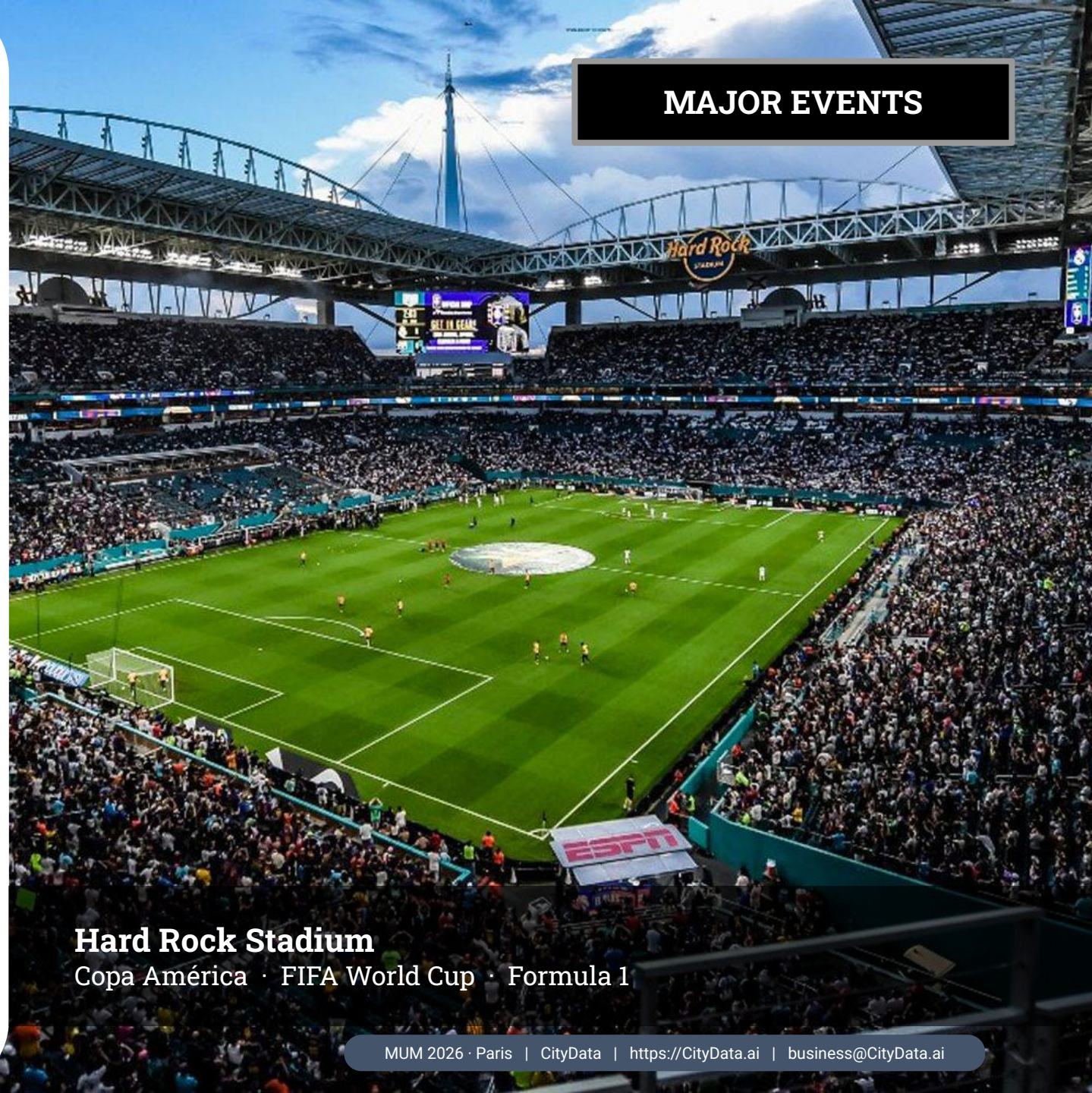
The network has to know when each movement gets a green light.

2 Event parking

DAY PLANS

The day plans have to know where people actually park.

MAJOR EVENTS



Hard Rock Stadium

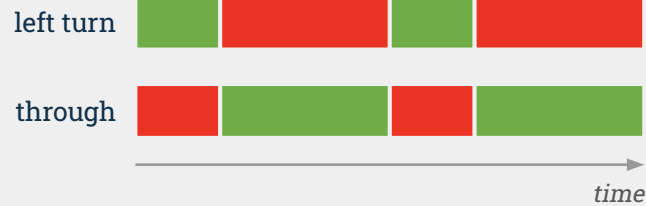
Copa América · FIFA World Cup · Formula 1

A Few Minutes of Green Time

MIAMI-DADE

Challenge 1 – the network: signal phases become scheduled capacity

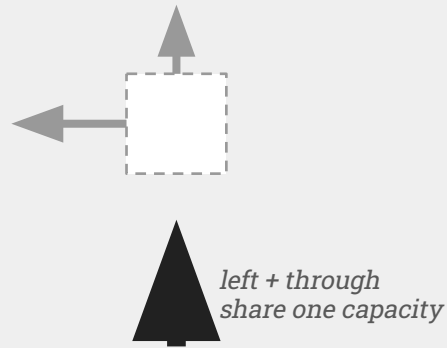
SIGNAL TIMING → CAPACITY



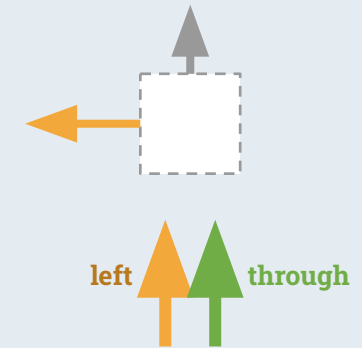
RED capacity → 0

GREEN capacity restored

BEFORE – one link, one capacity



AFTER – two links, two phases



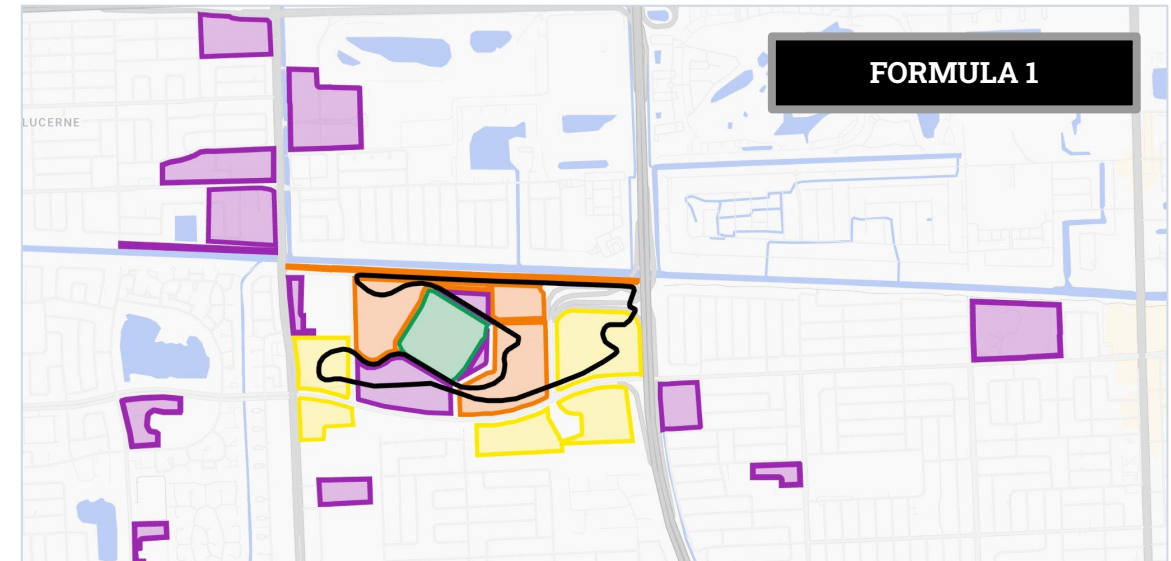
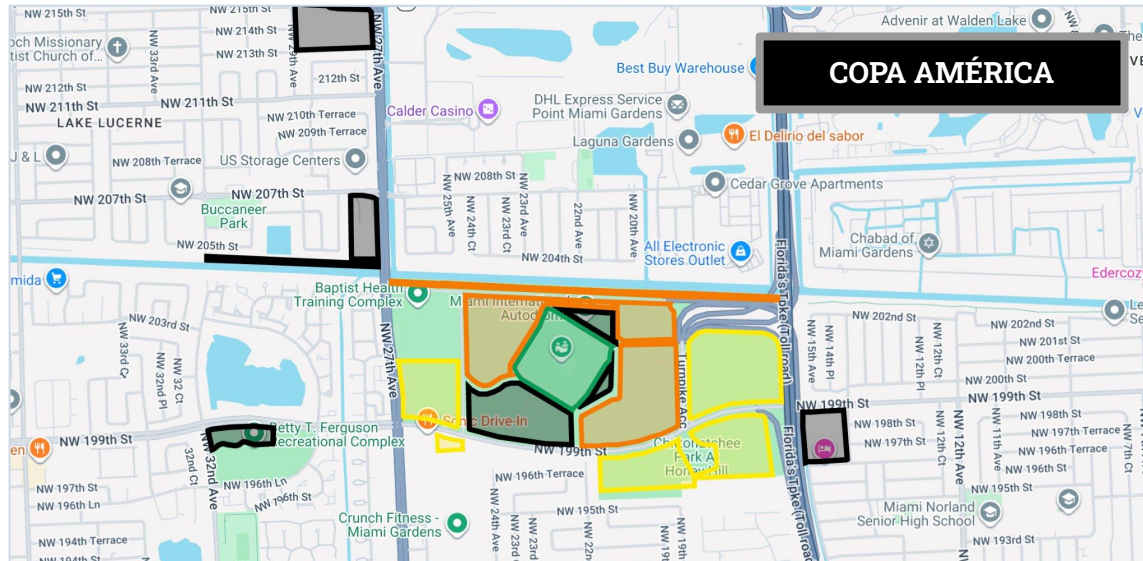
Gate a shared link and you stop both movements; leave it open and you stop neither.

A model doesn't necessarily tell you when it's wrong.

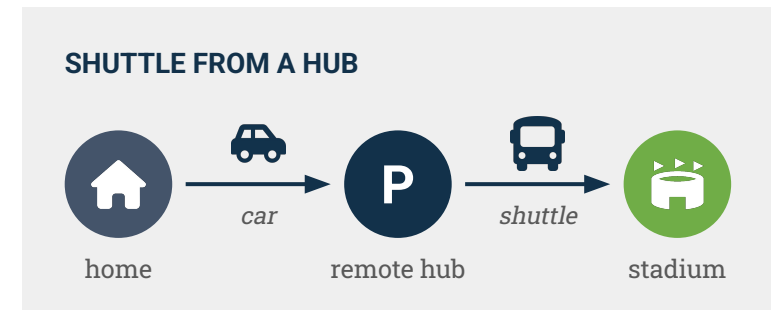
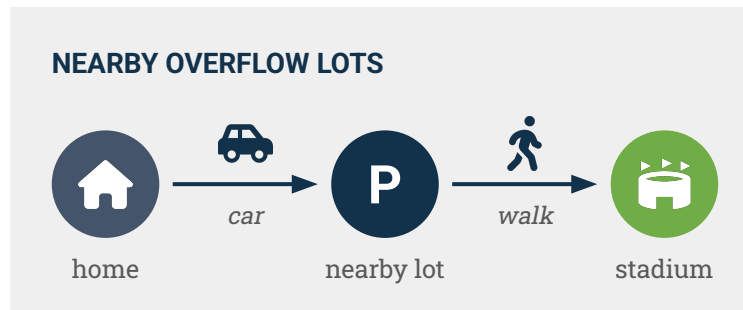
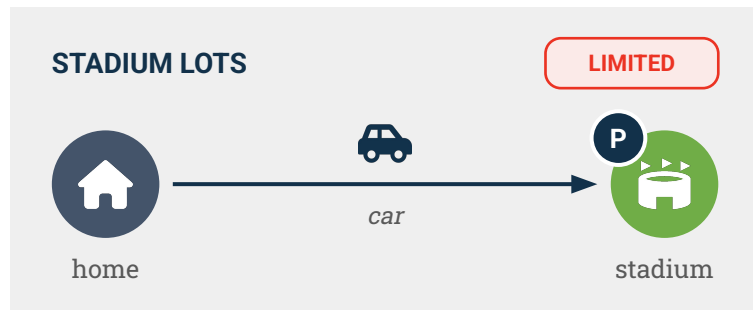
It can give a very precise answer built on an incorrect assumption. You have to look for results that don't make sense, and investigate them.

MIAMI-DADE

Challenge 2 – the day plans: every extra parking lot has to be in them



Hard Rock Stadium and the parking lots mapped for each event



CASE STUDY 3



Where We Are

A methods contribution, not a finding – yet

1 Traffic signals

IN THE MODEL

2 Event parking

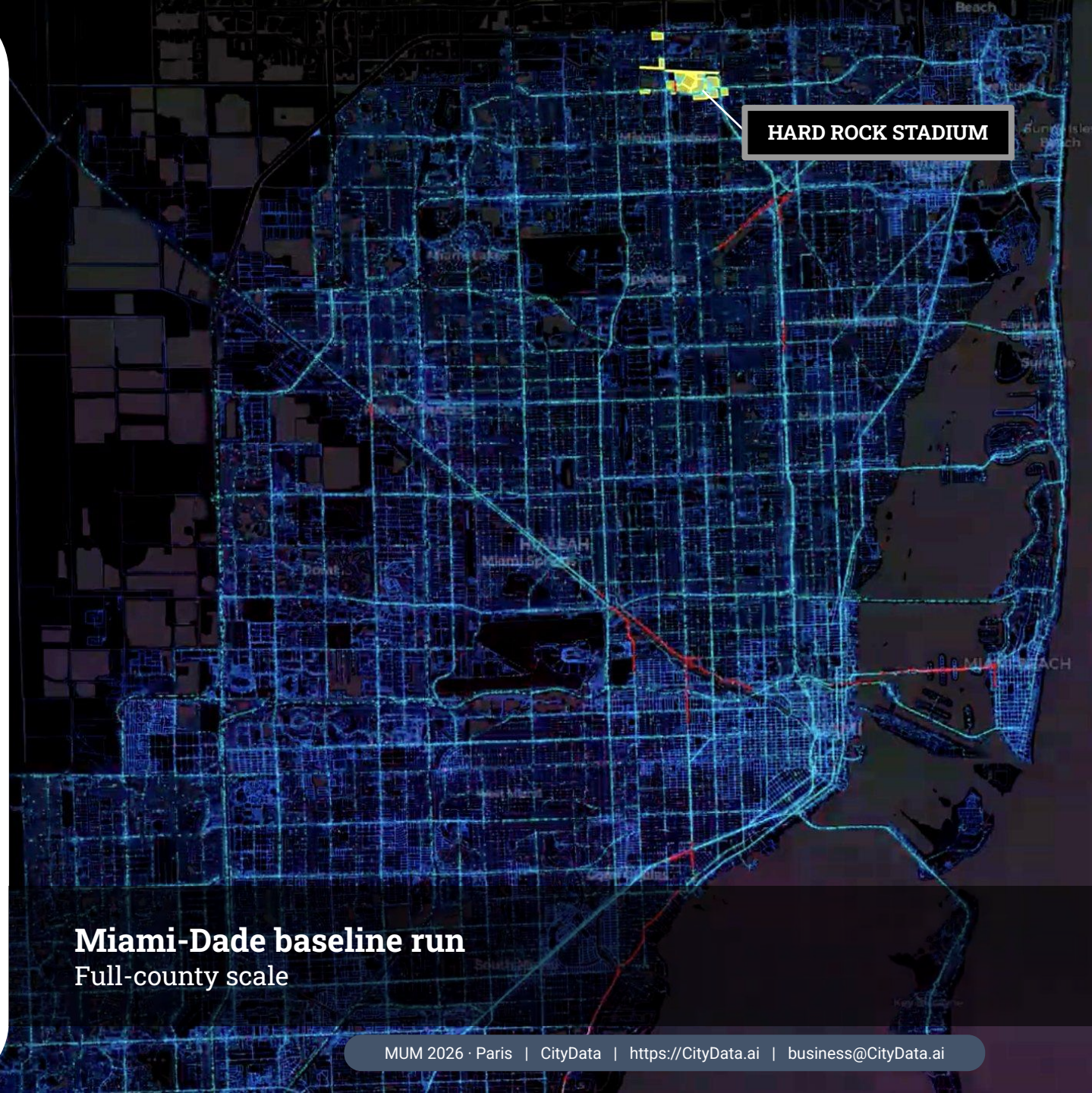
NEXT

Event scenario

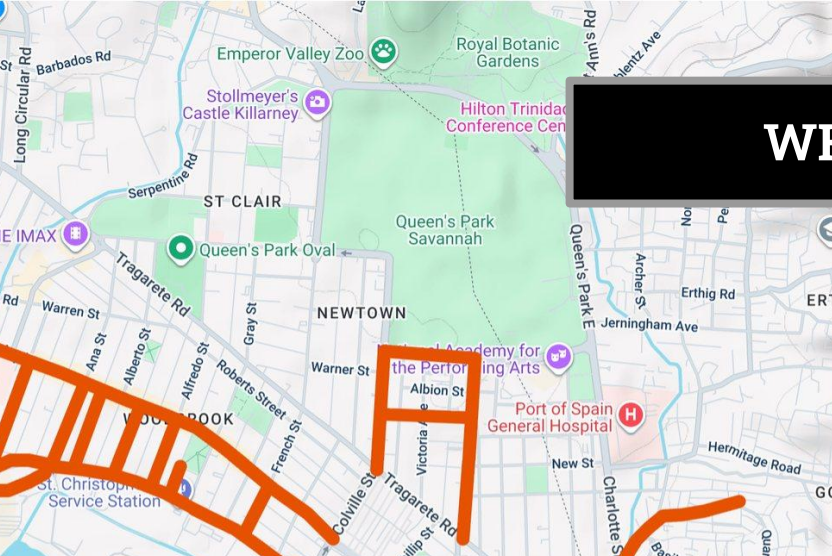
NOT YET

We hope to bring results next year.

Miami-Dade baseline run
Full-county scale



WHAT DO THESE THREE CITIES TELL US?



PORT OF SPAIN

Scenario-analysis tool

Change the network, run it again, and compare the outcomes.



VARANASI

Forecasting engine

Constrained by live observations, inside an estimation loop.



MIAMI-DADE

Extending what the network can represent

So we can evaluate an intervention that wasn't possible to model.

We're not claiming one simulation configuration solves every urban problem. We're demonstrating that a common data and simulation pipeline can support very different applications — without rebuilding the process from scratch each time.

From a two-metre-wide historic lane to hundreds of thousands of vehicles across a metropolitan road network.



What We Still Don't Know

A question that runs through all three projects:

How closely does the simulated city actually resemble the real one?

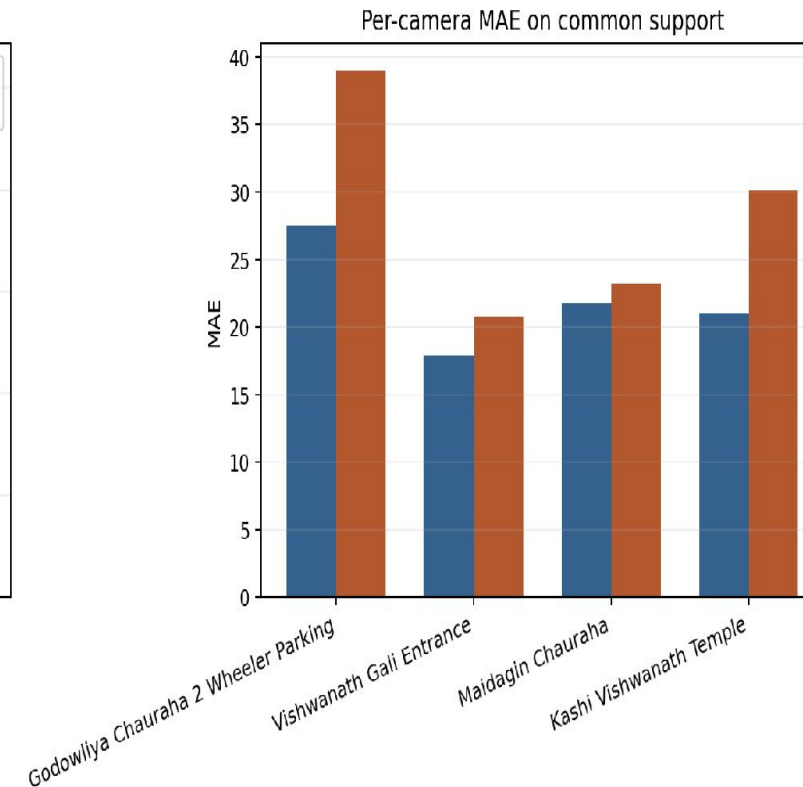
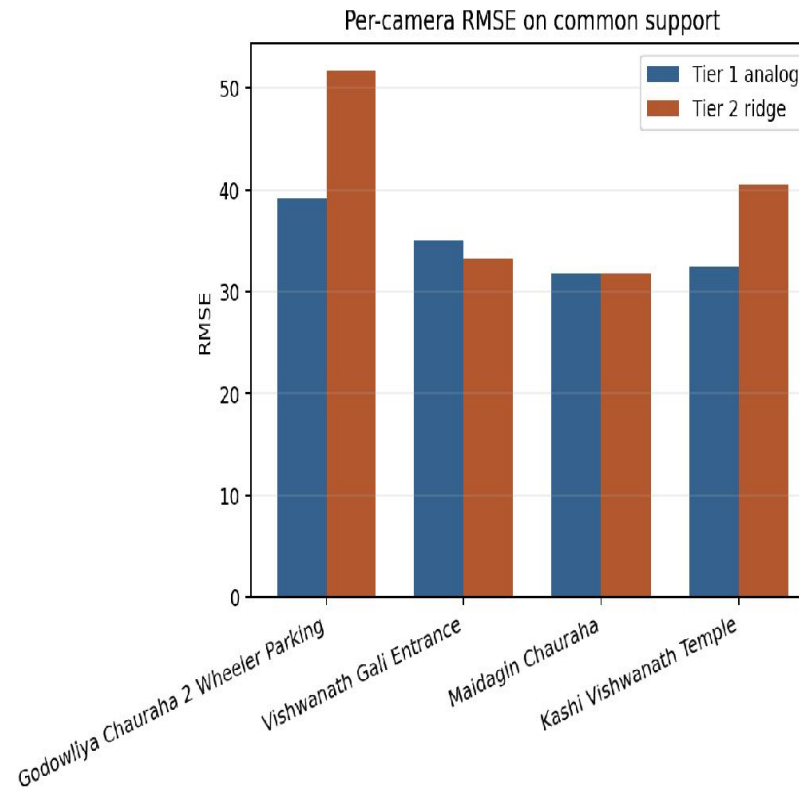
- 1 The validation problem**
Robust ground truth, and stronger ways to show our virtual population behaves like the real one.
- 2 The sensor-mapping gap**
Connecting what the simulation predicts with what individual sensors actually observe.

They're some of the most important questions for the next stage of the work.

One data pipeline, three cities, and a path from large-scale mobility data to a working laboratory for real-world urban decisions.

Varanasi – Per-Camera Comparison

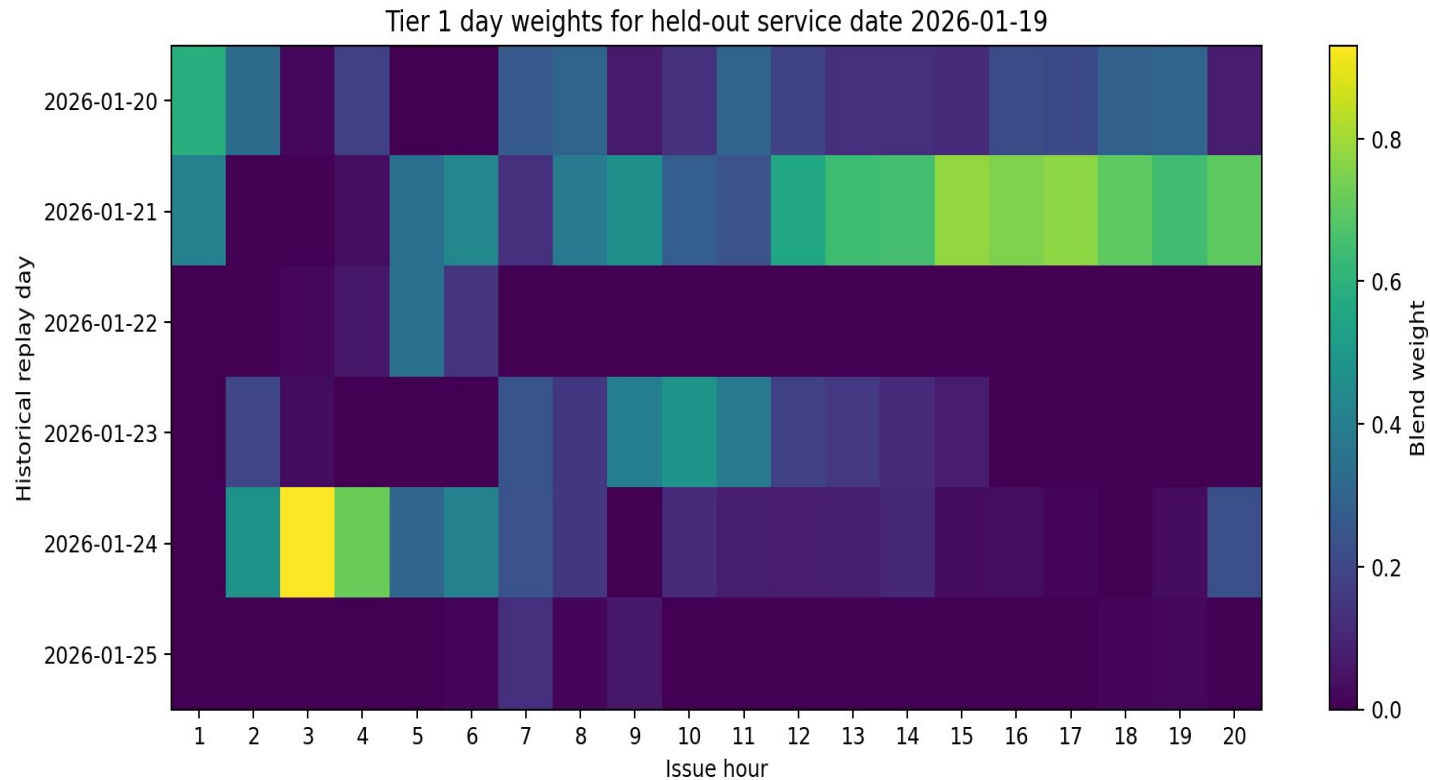
Tier 1 analog vs Tier 2 ridge, on common evaluation support



Tier 2 wins only on Vishwanath Gali RMSE and ties at Maidagin. Godowliya is where the gap is widest – and it is also the site with the largest `sim_inflow` coefficient mass in Tier 2.

Varanasi – Day Weights by Issue Hour

Held-out service date 2026-01-19; the target day is excluded from its own candidate library



Weights shift as the prefix lengthens and never collapse onto one row — 2026-01-24 dominates early, 2026-01-21 from midday on. This is the visible form of the “moderate mixture” claim.

Port of Spain – Key Performance Indicators

Baseline vs flooded scenario, trimmed mean (1st–99th percentile)

Table 1: Comparison of Key Performance Indicators (Baseline vs. Flooded Scenario)

Metric	Baseline Scenario	Flooded Scenario	Absolute Change	Percentage Change
Total Car Legs	265,397	256,319	-9,078	-3.4%
Mean Travel Distance (km)	26.43	26.74	+0.31	+1.2%
Mean Travel Time (min)	75.13	105.90	+30.77	+40.9%
Mean Speed (km/h)	22.30	20.58	-1.72	-7.7%

Data sourced from simulation output analysis.¹⁴

Port of Spain – MFD Capacity Analysis

Moving-window MFD from 5-minute leg-histogram data

Table 3: Network Capacity Analysis from MFD (Baseline vs. Flooded)

Metric	Baseline Scenario	Flooded Scenario	Percentage Change
Peak AM Arrival Rate (veh/h)	252,070	226,920	-9.98%
Peak PM Arrival Rate (veh/h)	200,680	189,990	-5.33%
Mid-Density Discharge Slope	0.704	0.460	-34.6%

Do not quote the absolute arrival rates aloud. They are labelled veh/h at ~252,000 for a demand level that under-captures reality by ~10×, and the aggregation window is not documented. The percentage deltas are the defensible claim.

Miami-Dade – What Has Actually Run

Baseline link volumes only. No change events, no intervention comparison.

What exists

- Baseline Miami-Dade runs on an OSM-derived network
- Copa América baseline link volumes (CSV + GeoJSON)
- Populations at 10 / 15 / 25 / 50 percent samples
- 24 signal assets parsed, mapped and verified end to end

What does not exist yet

- No networkChangeEvents applied to a scenario run
- No baseline-vs-intervention comparison
- No event-surge demand – the genuinely open problem
- 3,055 signals county-wide; 24 modelled

The event surge is the part that isn't sitting in historical passive data in the form you need. That is where I would most like this room's opinion.